AUTOMATION CASE STUDY

Pre-renewal grievance detection

Five n8n workflows connect lease timing, AI drafts, human approval, Gmail replies and visible issue tracking.

lease_id	status	touch_stage
cedar-row-9-2026-11-11	AT_RISK	CHECK_IN_90
cypress-4-2026-11-07	RESOLVED	CHECK_IN_90
ash-court-2d-2026-12-01	AWAITING_APPROVAL	CHECK_IN_90
linden-house-14-2026-11-19	AWAITING_APPROVAL	CHECK_IN_90
canyon-ridge-15-2026-12-...	AWAITING_APPROVAL	CHECK_IN_90
elm-terrace-6-2026-11-23	AWAITING_APPROVAL	CHECK_IN_90
riverside-5b-2026-11-15	AWAITING_APPROVAL	CHECK_IN_90
harbor-flats-11-2026-11-28	AWAITING_APPROVAL	CHECK_IN_90
juniper-1b-2026-11-26	AWAITING_APPROVAL	CHECK_IN_90
parkside-tower-unit-8a-20...	AWAITING_APPROVAL	CHECK_IN_90
brookline-3d-2026-11-13	AWAITING_APPROVAL	CHECK_IN_90
summit-10c-2026-11-21	AWAITING_APPROVAL	CHECK_IN_90
willow-court-5b-2026-11-21	AWAITING_APPROVAL	CHECK_IN_90
vista-16-2026-11-19	AWAITING_APPROVAL	CHECK_IN_90

Actual Airtable records from the preserved synthetic demo.

The problem

An expiry list says when a lease ends. A manager still needs a way to ask about unresolved concerns and follow through.

What I built

A manager-controlled check-in and reply loop using n8n, Airtable, Telegram, Gmail and OpenAI.

Demonstrated outcome

14 drafts prepared, 2 check-ins delivered, 2 replies routed and 1 issue marked resolved by the manager.

Ask early. Put the answer in front of a person.

The build turns a lease roster into a repeatable check-in, review and issue-follow-up process.

Before a check-in

Lease dates are available, but preparing personalized messages, following replies and updating issue status takes several manual steps.

The working loop

Date rules select the outreach window. OpenAI prepares wording. A manager reviews each message in Telegram before Gmail sends it.

After a reply

The automation matches the resident, records the reply and routes concerns with their original wording and a suggested human action.

The manager stays in control

Buttons support editing, individual approval, skipping, reply assistance and issue closure. A reply draft needs its own send approval.

Visible state

Airtable holds resident status, draft wording, reference codes, replies, issue fields, an event log and summary metrics.

The sample you are viewing

Screenshots show real applications and preserved executions from August 2026. Resident identities, properties and reply scenarios are synthetic.

Demo boundaries

Gmail, Telegram, OpenAI and Airtable are genuine integrations. A local n8n instance orchestrates them; roster and reply scenarios are demo fixtures.

One connected automation

Airtable stores state. n8n moves the data. Telegram is the manager's action surface.

01 **Outreach generation / WF7**

Selected lease window → OpenAI draft → wording checks → Airtable draft → Telegram review card

02 **Manager actions / WF6**

Telegram button or command → current Airtable record → individual approval → Gmail → event log

03 **Reply processing / WF8**

Gmail reply → reference match → routing or classification → Airtable issue → manager alert

04 **Operations / WF9**

Daily schedule or refresh → date cohorts, silence escalation and metrics → Airtable / Telegram digest

05 **Automatic checks / WF10**

Minute timer → silent call to WF8 → new replies processed through the same matching rules

Shared data

Four Airtable tables hold leases, outreach events, configuration and metrics. The model classifies reply text; resident identity comes from deterministic matching.

Lease dates drive the outreach window

The demo contains 40 synthetic residents. Timing groups are computed from the lease end date.

lease_end_date	days_to_expiry	cohort
2026-11-11	80	90
2026-11-07	76	90
2026-12-01	100	90
2026-11-19	88	90
2026-12-04	103	90
2026-11-23	92	90

Actual input fields

Lease end date, days until expiry and cohort are shown in the populated Airtable table.

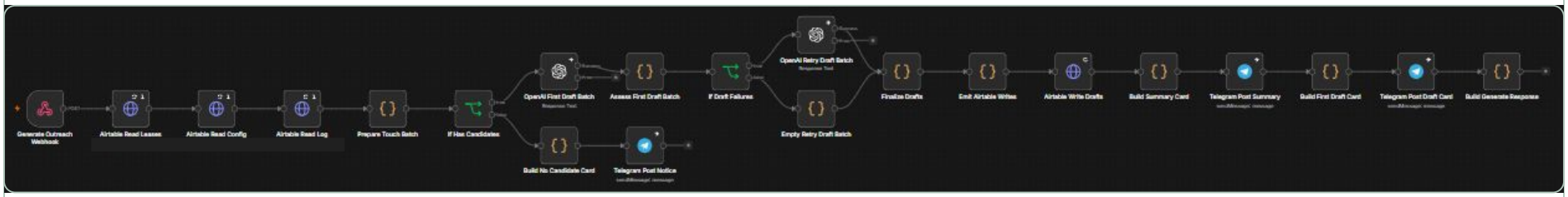
Same-record journey

The first two rows belong to Avery Brooks at Cedar Row 9 and José Núñez at Cypress 4. Their messages and results appear throughout this sample.

- The recorded seed produced 14 residents in the 90-day group.
- Verified roster checks flagged 8 messy rows for attention rather than dropping them.
- 120-day records are a watchlist; 90-day check-ins and eligible 60-day nudges produce reviewable drafts. The 30-day stage is a human handoff.

Prepare the check-in queue

Actual n8n editor capture after Tidy Up. All 21 nodes are visible; enlarged branches follow.



FULL WORKFLOW / DARK THEME

Trigger and input

A generation request starts the workflow, normally through a manager command. It reads leases, configuration and prior outreach from Airtable.

Key steps and output

Date rules select eligible residents; OpenAI drafts wording; fixed checks allow one retry. Usable drafts are stored as awaiting approval, and Telegram receives a batch summary and the first review card.

Select records before asking the model

Native-zoom captures of the input chain and the candidate decision.



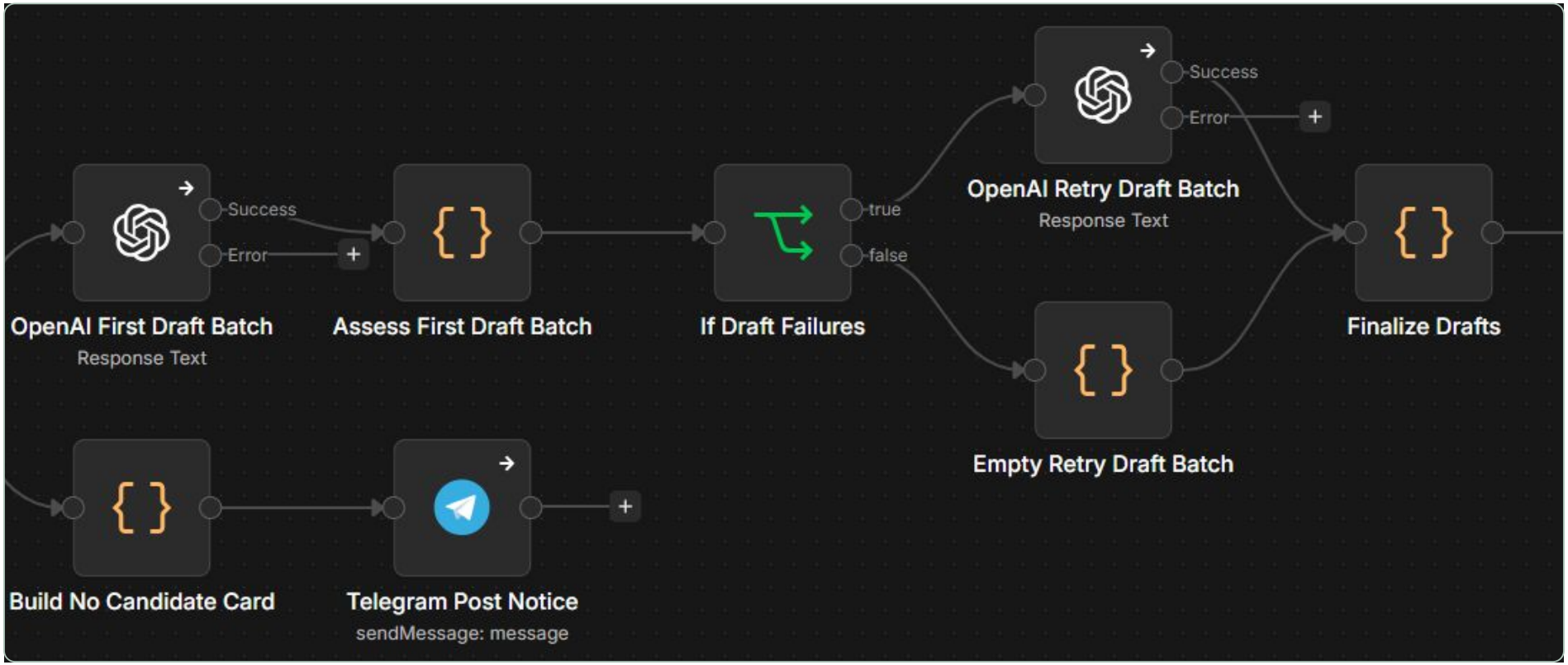
The request enters n8n. Lease rows and settings come from Airtable.



Prior sends and opt-outs are checked. Prepare Touch Batch selects the date-based group; If Has Candidates routes the result.

A draft must pass the wording checks

The OpenAI branch and retry decision are captured directly at native editor zoom.



One retry, then visible review

Only failed wording goes to the retry model. A draft that still fails remains for manager wording.

Empty group

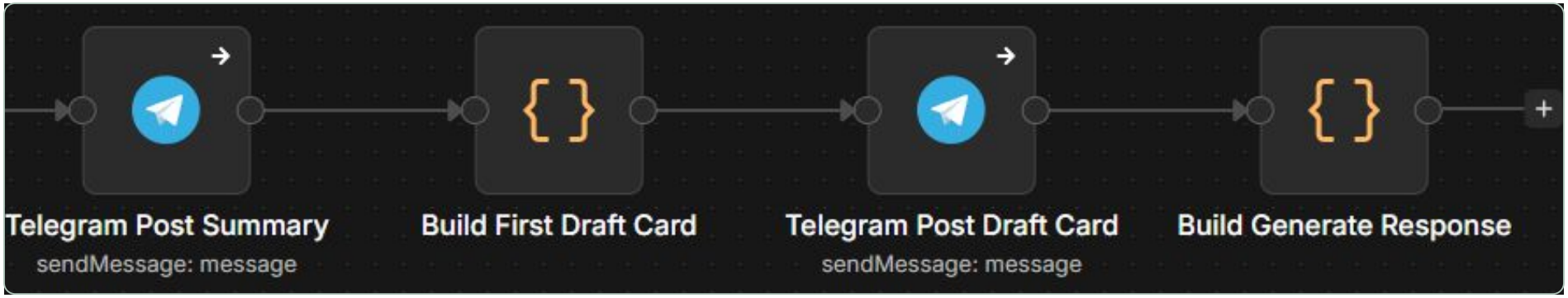
The lower branch builds a no-candidate notice instead of calling the drafting model.

Save the wording, then open the review

Draft generation prepares the queue. Individual delivery happens in the manager-actions workflow.



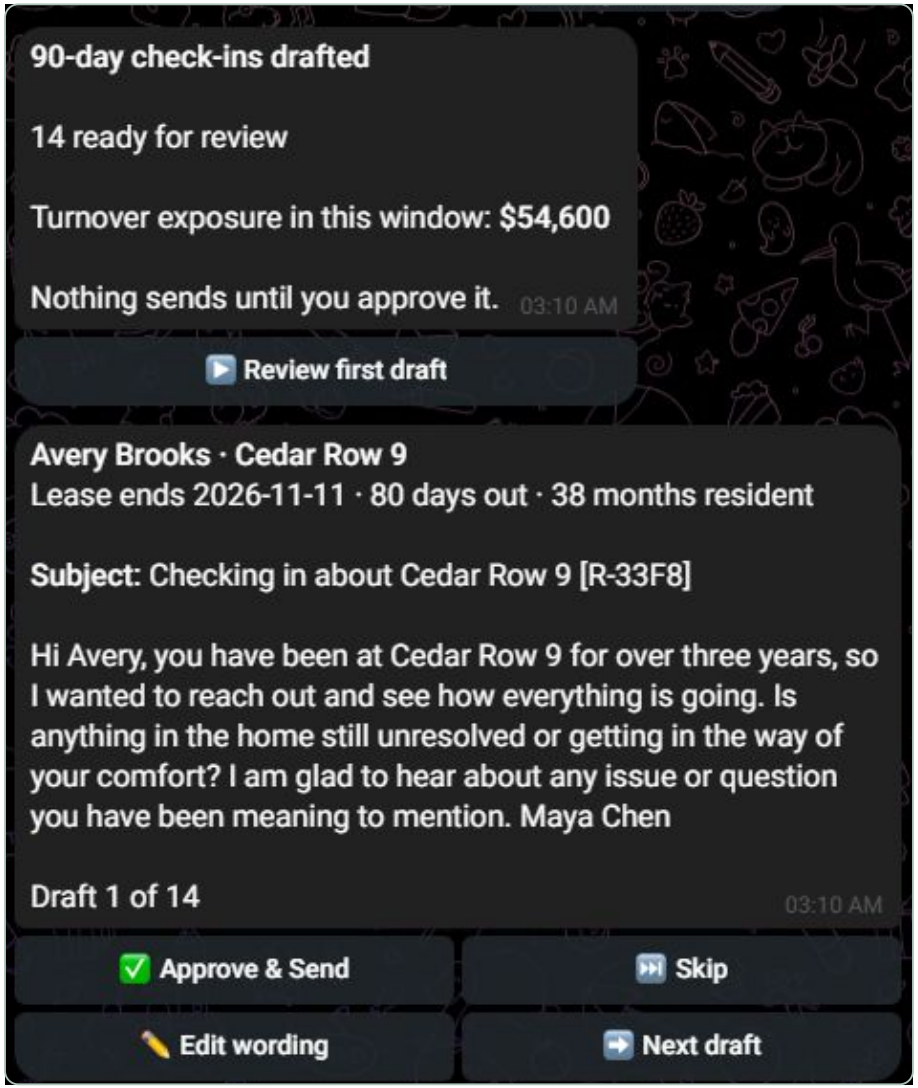
Finalize Drafts prepares lease updates and audit events. Writes are batched in groups of up to ten records.

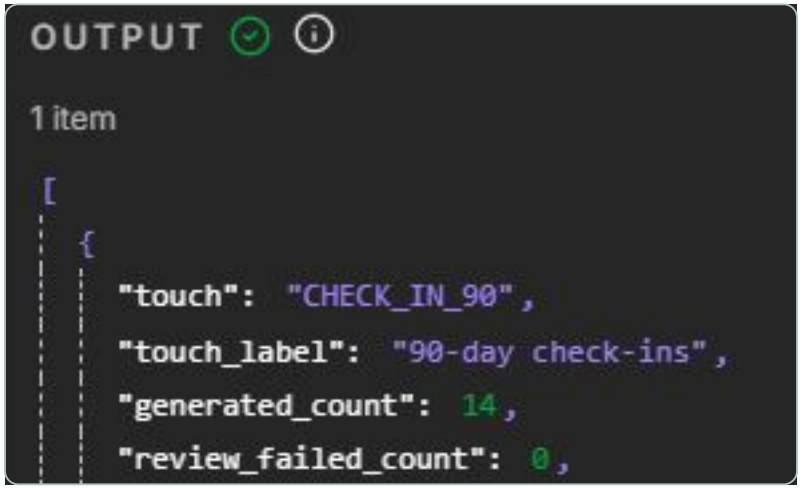


Telegram posts the batch summary and first draft card. The response reports what generation produced.

Fourteen drafts became a review queue

Preserved execution and conversation from August 24, 2026. All resident data shown is synthetic.





Execution result

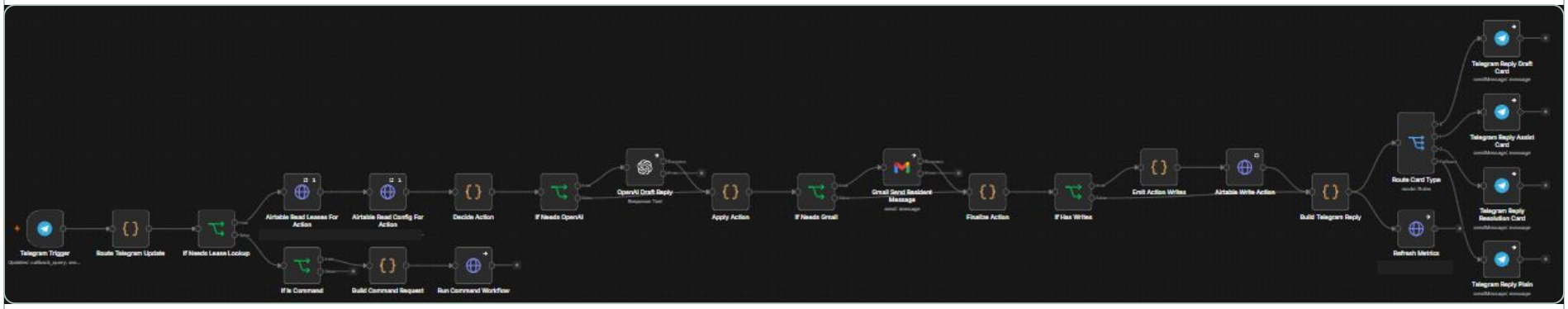
The recorded run completed in 24.373 seconds. Finalize Drafts returned generated_count 14 and review_failed_count 0.

Business output

Telegram shows 14 ready for review and Avery's first draft with reference R-33F8. The buttons offer approval, skip, editing and the next draft.

One entry point for every human action

Actual n8n editor capture after Tidy Up. All 25 nodes are visible; enlarged branches follow.



FULL WORKFLOW / DARK THEME

Trigger and input

The project's only Telegram Trigger receives buttons and slash commands. It reads the current lease and configuration before applying a resident action.

Key steps and output

The router separates commands, edits, approvals, assistance and issue closure. An approved send passes the Gmail gate, updates Airtable and returns a relevant Telegram card.

One trigger receives the manager's actions

Native-zoom capture of Telegram intake and the first routing decision.



Trigger and data

Telegram Trigger receives button callbacks and plain messages. Route Telegram Update turns them into a known action or command.

Decision

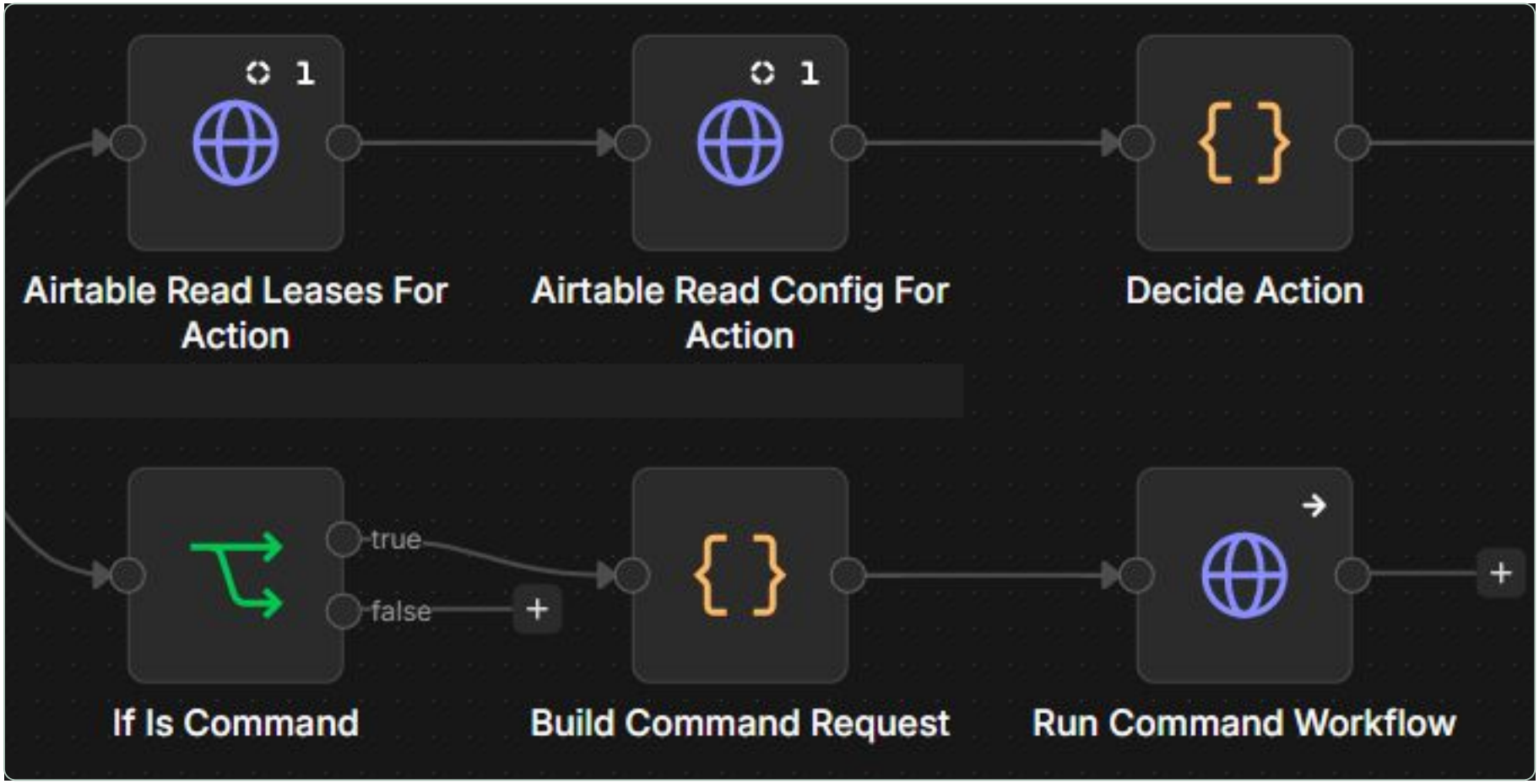
If Needs Lease Lookup determines whether the action requires a resident record. A button press carries the record context; manager commands dispatch the relevant operation.

Human intervention

Editing, individual approval, reply assistance and resolution all enter through this trigger. The next page shows the current-state lookup and command paths.

Read current state before applying the tap

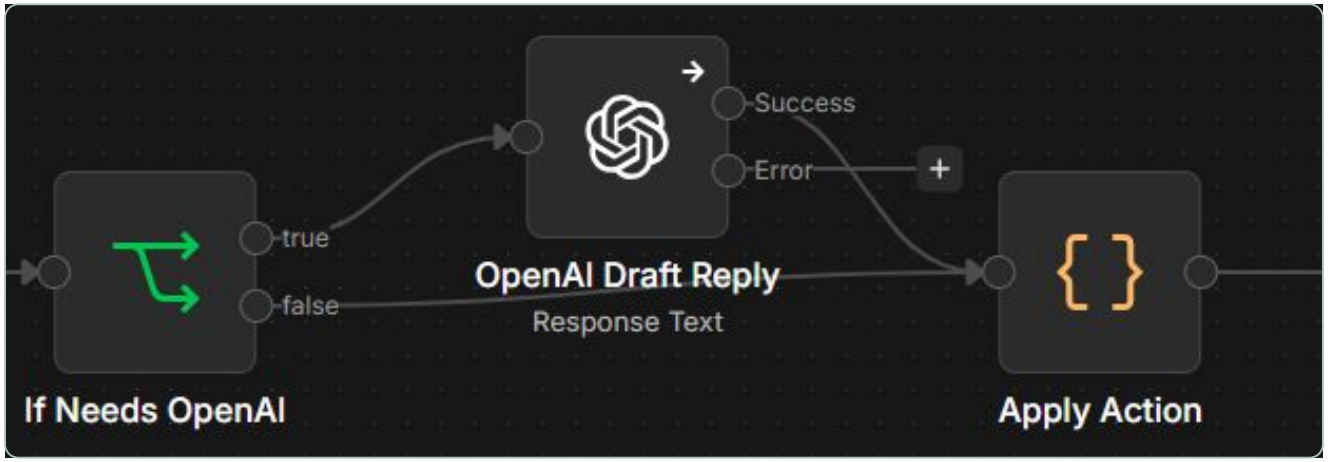
The upper branch handles resident actions. The lower branch dispatches manager commands.



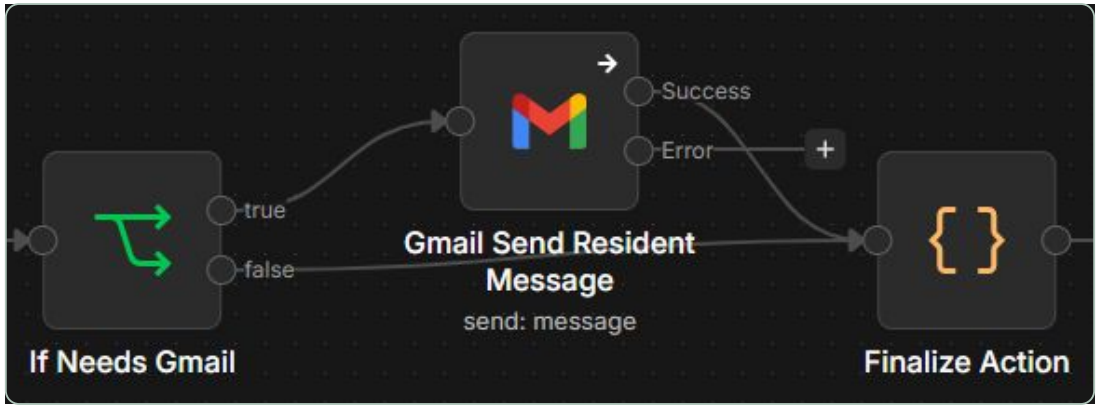
Resident actions read Airtable and reach Decide Action. Commands become a request to the relevant workflow. Current records are read again before send-time checks are applied.

Only an approved action reaches Gmail

Native-zoom captures make both decisions and node labels readable.



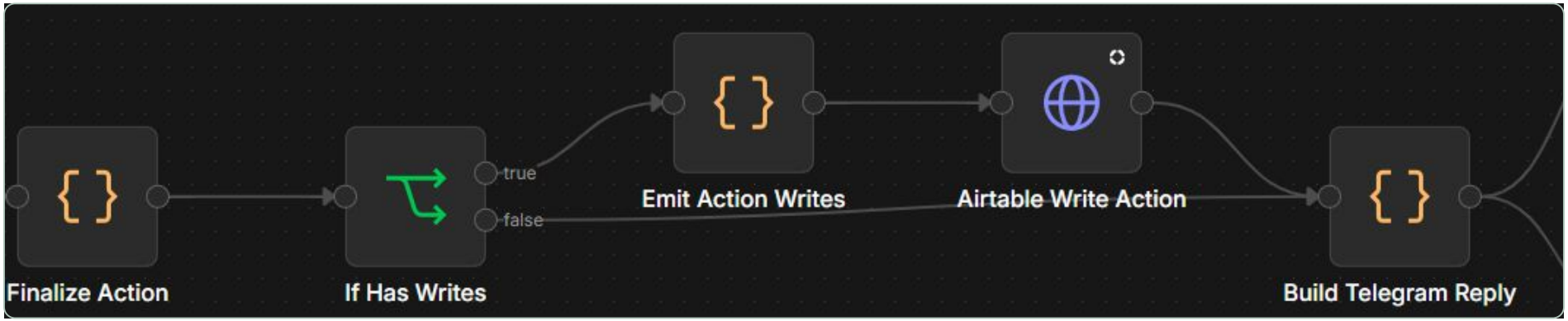
Reply assistance can call OpenAI. Apply Action checks the action and wording; a draft is still unsent.



If Needs Gmail opens the send path only for the appropriate approved action. Finalize Action reads the send result before preparing updates.

Record the result after the action

The workflow keeps message delivery and the manager's visible state connected.



State changes

Finalize Action prepares the lease update, any pending edit state and the outreach event. If Has Writes routes only the needed changes.

Delivery evidence

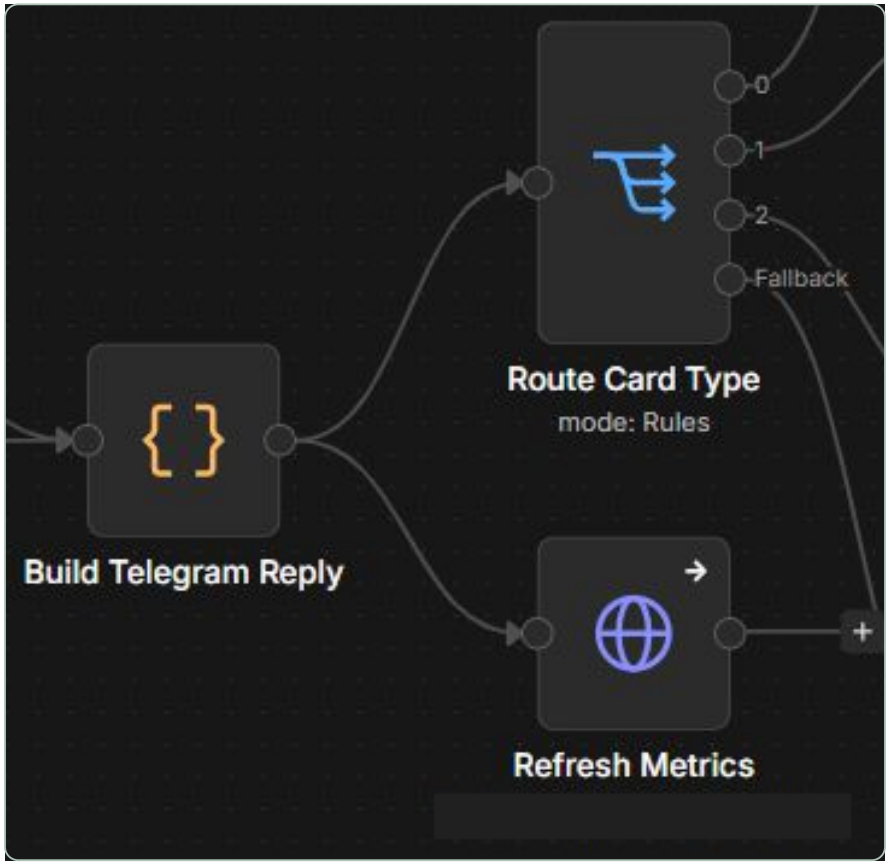
Successful Gmail sends contribute their message identifier to the audit event. A failed send becomes a visible failed status and manager response.

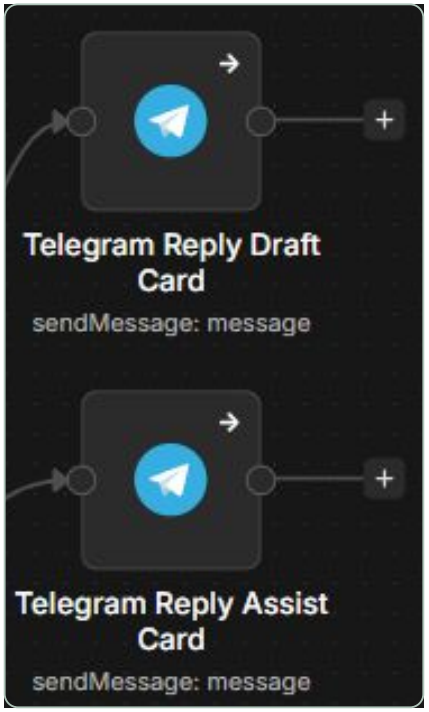
Next interaction

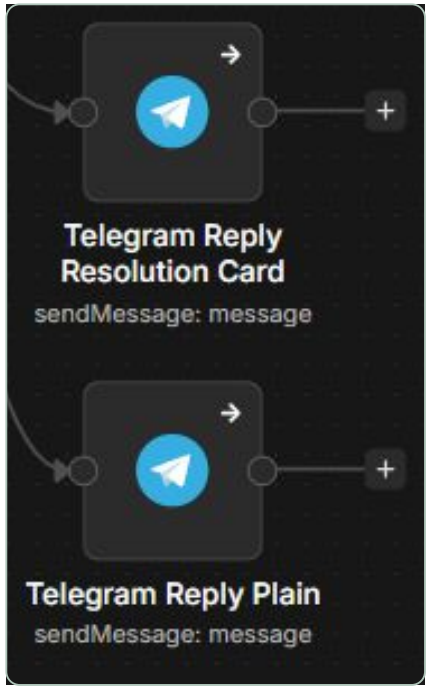
After Airtable writes, Build Telegram Reply prepares the next card. Metrics are refreshed from the resulting state.

Return the right controls for the next step

These are separate actual closeups of the routing branch and its four card outputs.



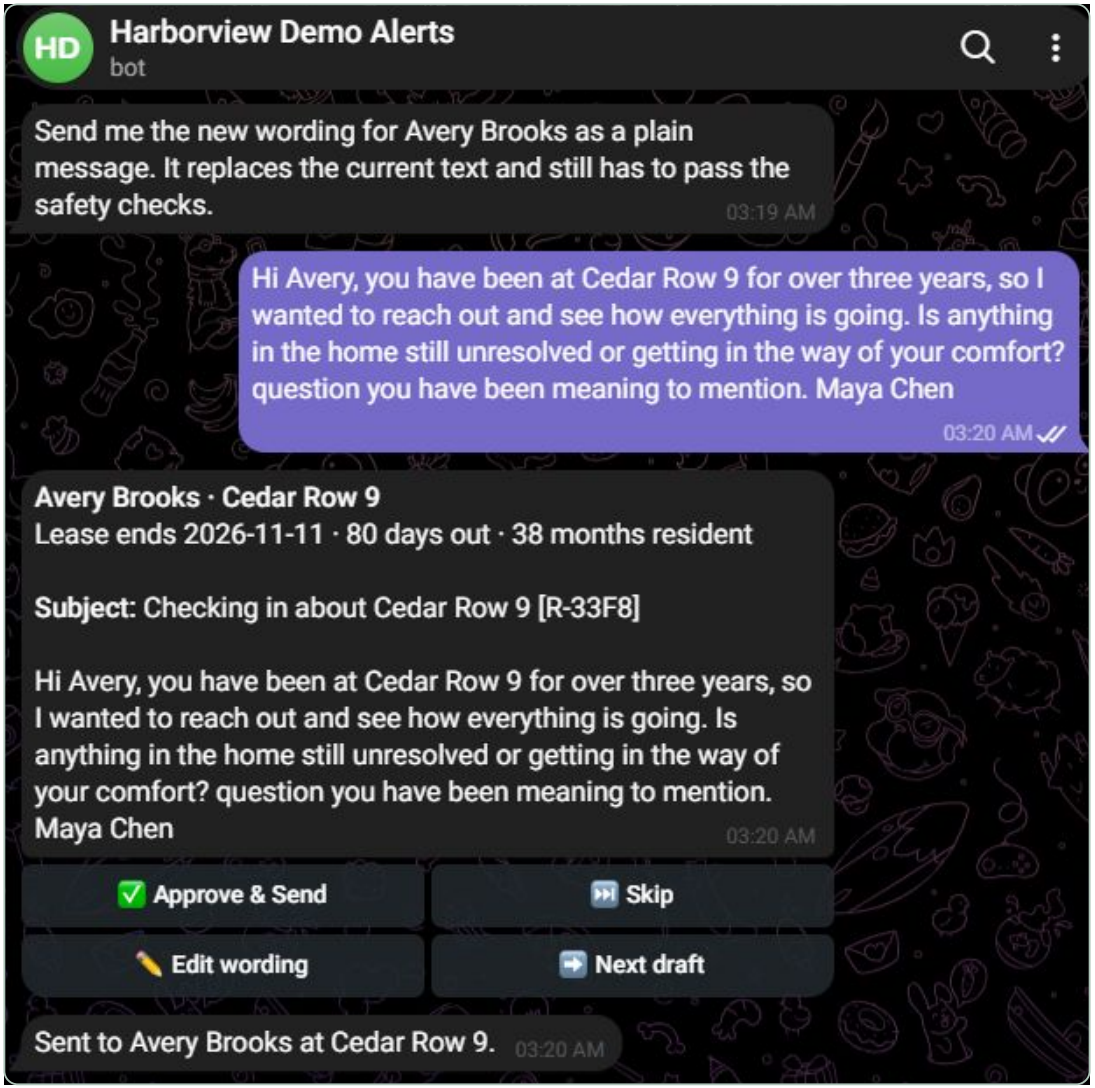




Route Card Type selects a draft, suggested reply, resolution or plain response. Each card gives the manager the controls appropriate to that state.

The manager edits, then approves Avery's message

Actual Telegram conversation / synthetic Avery Brooks / reference R-33F8.



Edit in context

The conversation shows the edit request, the manager's replacement text and a refreshed review card.

Individual approval

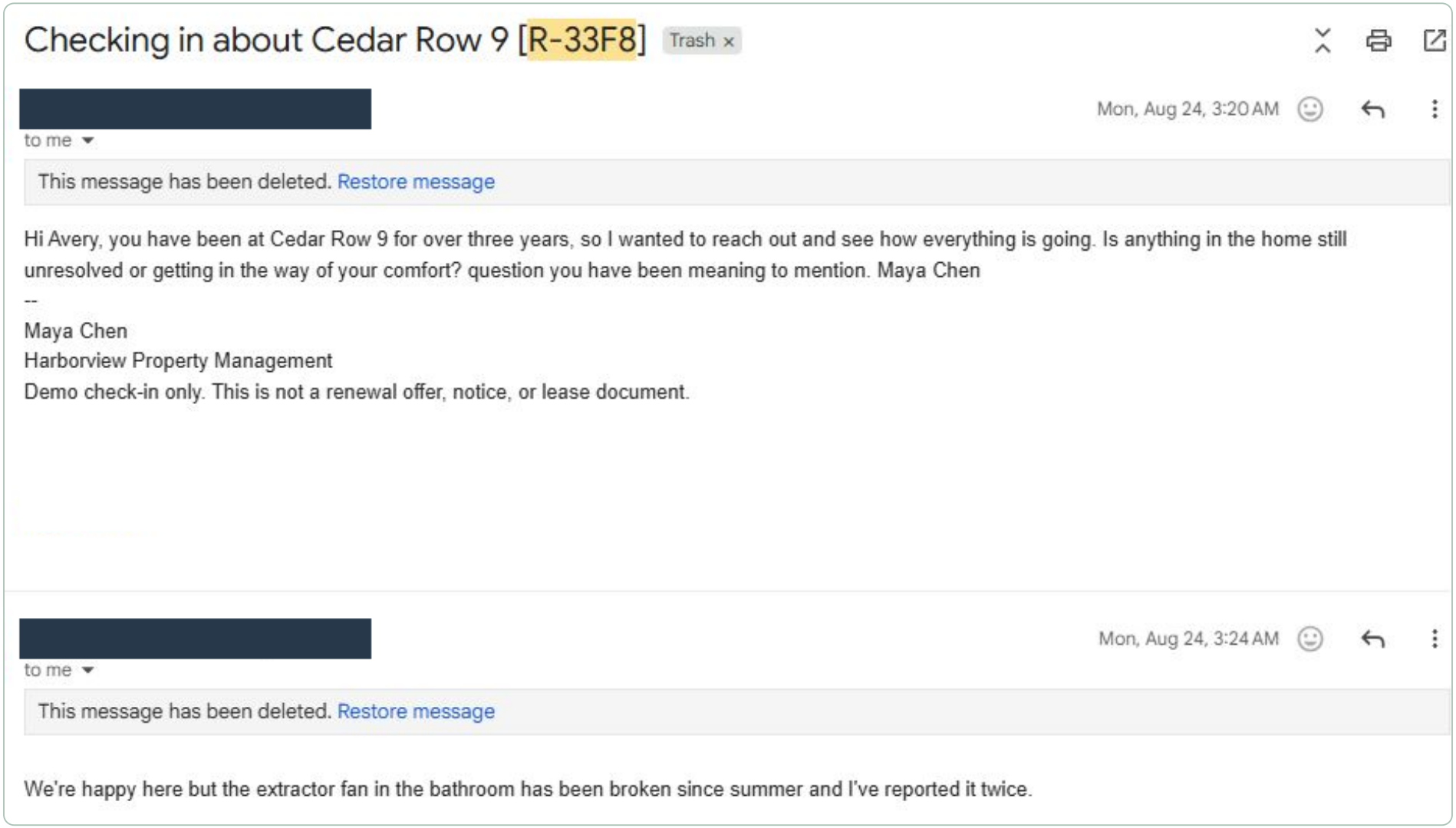
Approve & Send applies to Avery's message. The next response confirms it was sent to Cedar Row 9.

Traceable handoff

R-33F8 stays in the subject so the subsequent Gmail reply can be matched to the same record.

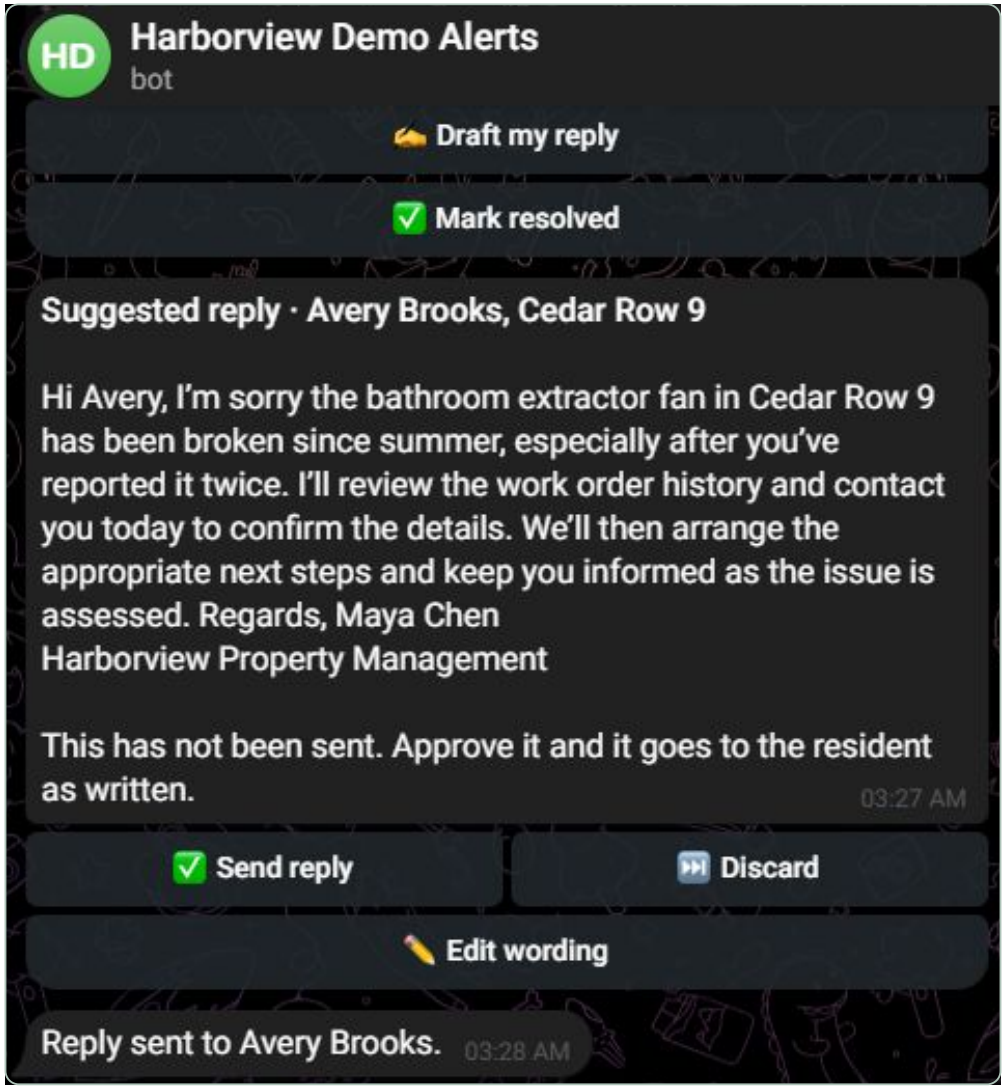
The approved check-in reached Gmail

Actual Gmail thread / R-33F8: edited check-in at 03:20, synthetic reply at 03:24. Contact headers redacted.



Suggested wording stays unsent until approved

Actual Telegram progression from “Draft my reply” to a send confirmation for Avery.



Assistance on request

The manager asks for a reply. OpenAI drafts a response using Avery’s maintenance concern.

Review before sending

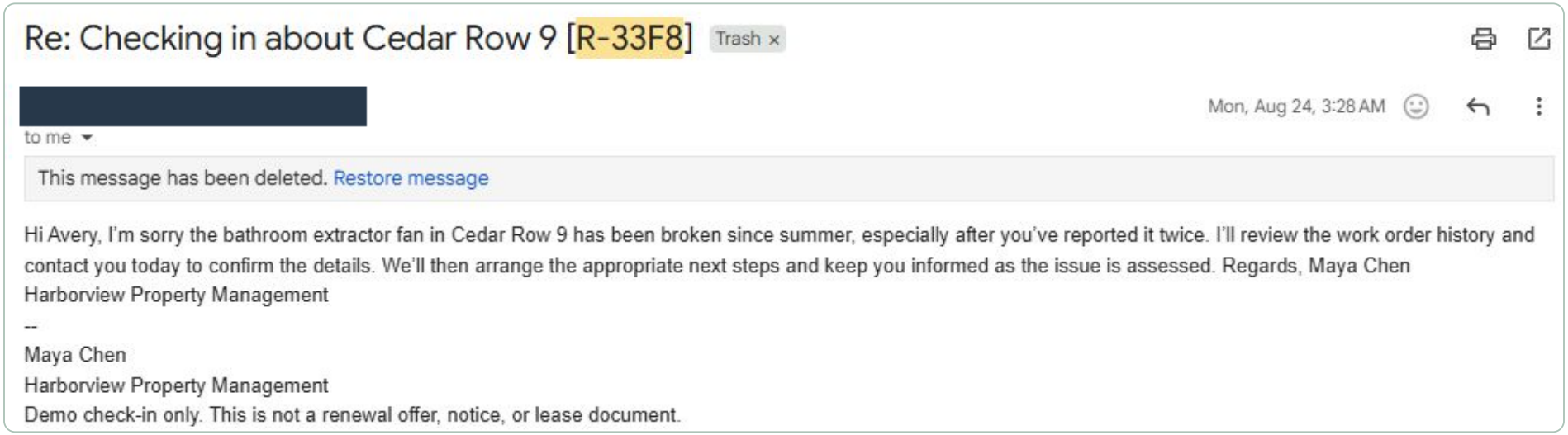
The card states that the wording has not been sent. It offers Send reply, Discard and Edit wording.

Confirmed action

The next conversation message says “Reply sent to Avery Brooks.” The delivered email appears on the next page.

The approved response also reached Gmail

Actual Gmail message / R-33F8 / 03:28. The body matches the suggested reply shown in Telegram.



Same subject, same issue

The reference code links the approved response to Avery's original check-in and maintenance reply.

Observable output

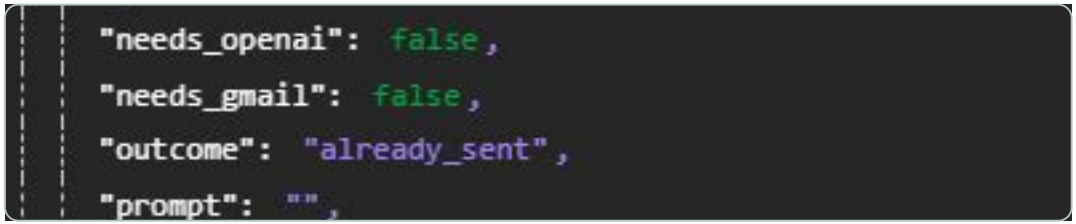
The email contains the reviewed wording about the extractor fan and next steps. The outreach event is recorded as ASSIST_SENT.

Synthetic demonstration

Both sides of the email exchange use the demo mailbox. The reply content is a synthetic resident scenario transported through genuine Gmail integration.

A second approval is refused

Separate recorded verification from August 21, 2026; this is not a second send in the R-33F8 journey.



Actual decision output

Decide Action returned outcome already_sent, needs_gmail false and needs_openai false. The repeated approval did not enter the Gmail branch.

Verified related controls

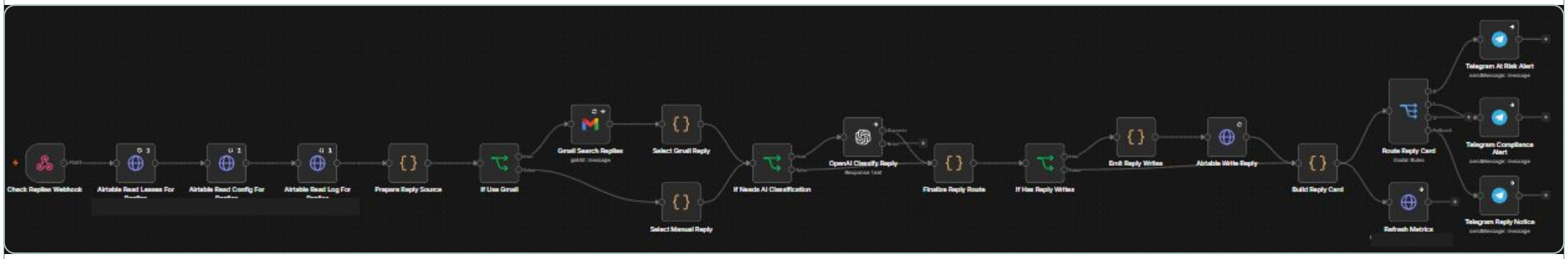
The project also verified that Skip refuses a resident already sent to, and that edited wording must pass the same checks as generated wording.

Why this matters

The recorded result demonstrates the refusal path as well as the successful delivery path. A successful workflow status alone would not show that distinction.

Match the resident, then route the reply

Actual n8n editor capture after Tidy Up. All 21 nodes are visible; enlarged branches follow.



FULL WORKFLOW / DARK THEME

Trigger and input

A manual check or the automatic timer calls the reply-check webhook. Airtable supplies leases, settings and previously logged message identifiers.

Key steps and output

Gmail provides new reply candidates. Reference codes match first; remaining text follows routing or structured classification. The workflow writes reply and issue fields, logs the event, refreshes metrics and posts the appropriate manager alert.

Load the context before reading replies

Native editor captures of the request and source-selection steps.



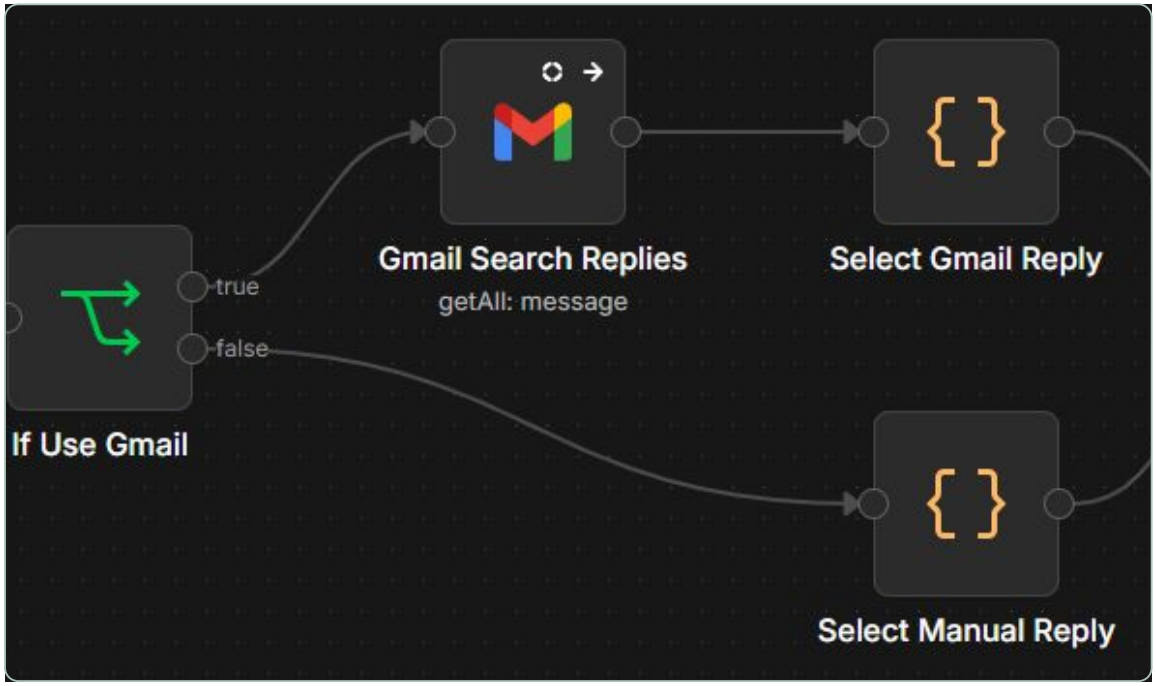
Check Replies reads the lease roster and configuration so every reply can be handled in context.



Prior message identifiers are read from the log. Prepare Reply Source chooses genuine Gmail intake or a clearly identified manual fixture path.

One matching process for both input paths

The enlarged branch shows Gmail intake and the manual scenario input used for controlled verification.



Genuine Gmail intake

Gmail Search Replies retrieves candidate messages. Select Gmail Reply ignores outbound check-ins and already logged messages.

Deterministic identity

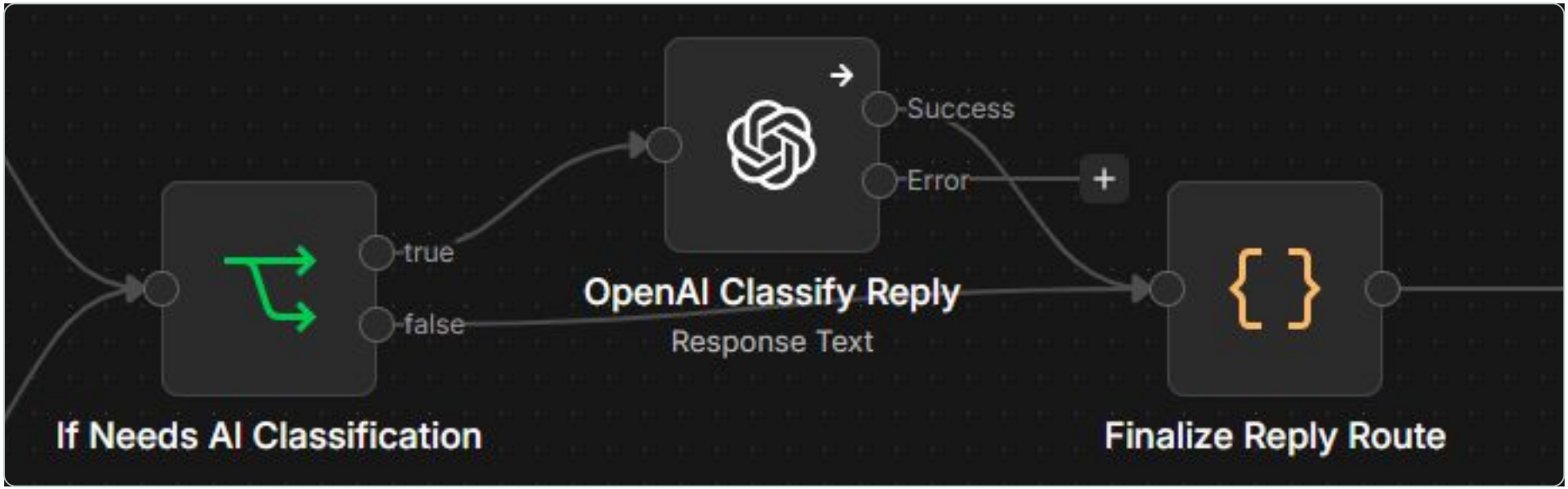
A subject reference such as R-33F8 matches first. Sender matching is a fallback; the model never chooses the resident.

Manual scenarios

Select Manual Reply supports fixture-based checks. Those checks are separate from the genuine Gmail journey shown here.

Route sensitive signals before the model

Actual native-zoom decision branch. The automation preserves the original reply for the manager.



Deterministic routes

Sensitive signals, opt-outs, no-signal cases, duplicates and unmatched messages can finish without a model call.

Structured classification

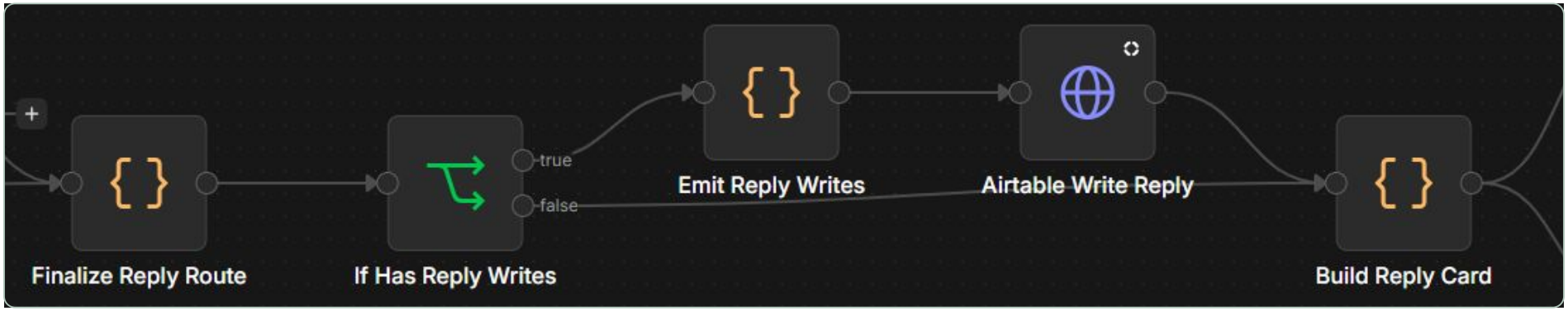
Other matched replies enter OpenAI. The result supplies intent, category, urgency, confidence and a suggested human action.

Human intervention

Only high-confidence staying replies without a grievance, compliance flag or question qualify for the clean-staying route. Other signals go to a person.

Persist the decision and prepare the alert

Native-zoom capture of the write branch after reply routing.



Lease and log updates

Finalize Reply Route prepares the resident's reply, intent, issue fields and outreach event. Emit Reply Writes sends the needed updates to Airtable.

Manager output

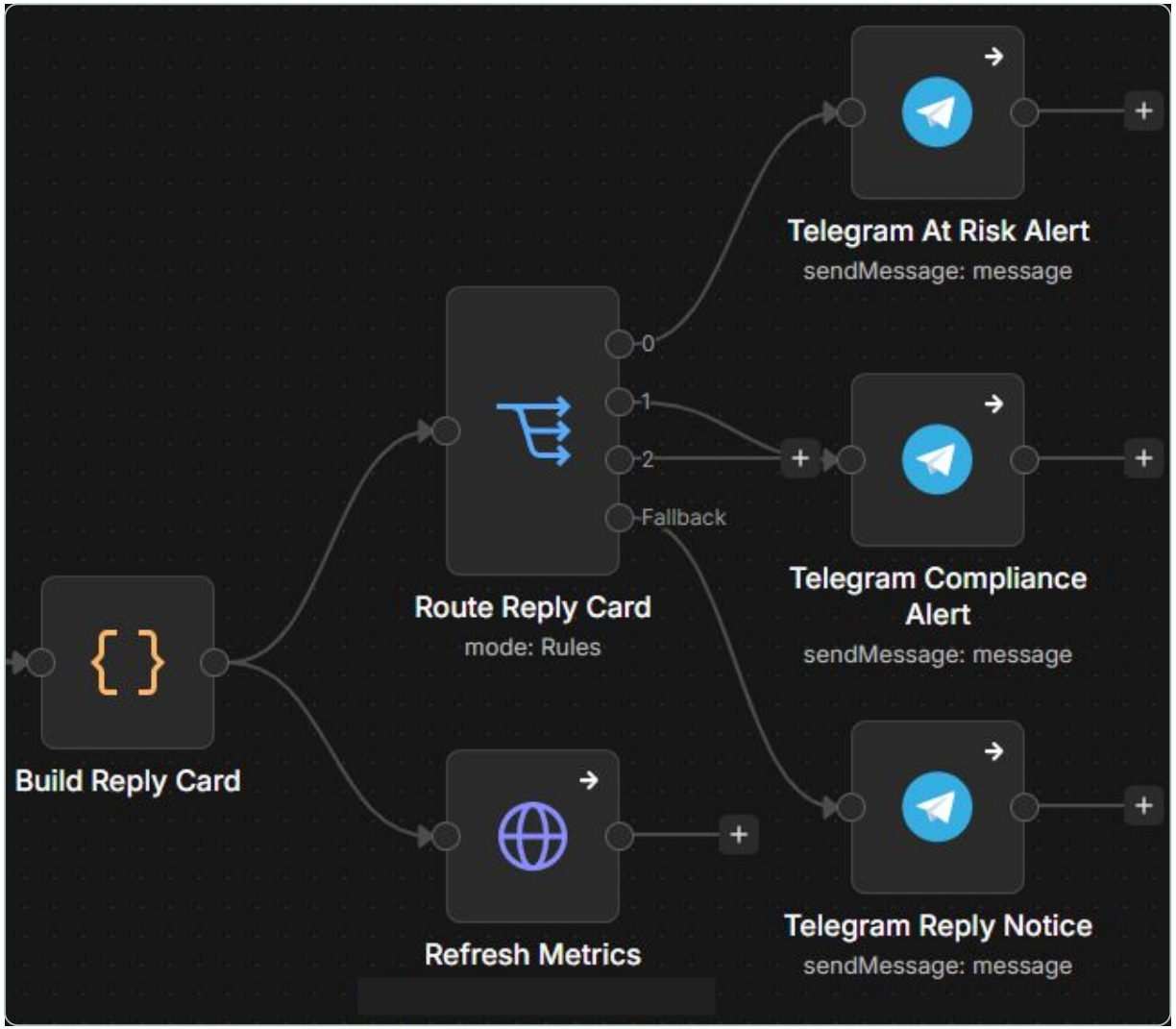
Build Reply Card prepares the quote, classification and next action. The alert type reflects the reply decision.

Visible failure handling

When classification is unavailable or unusable, the reply remains a human-review item rather than disappearing from the process.

Put the useful context on the manager's phone

Actual routing closeup. Metrics refresh follows the same recorded state change.



At-risk alert

A concern is shown with the original wording, resident context and a suggested action.

Sensitive alert

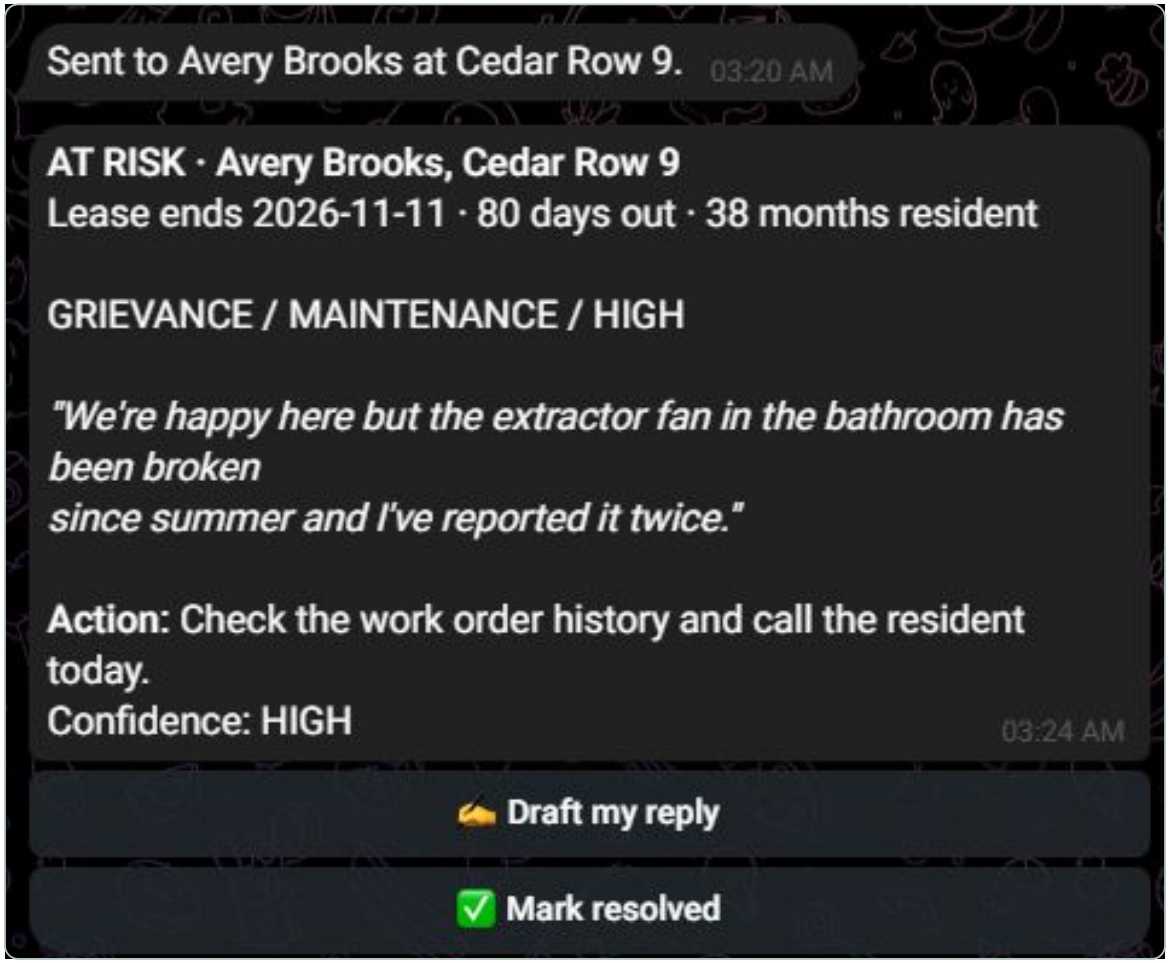
Accommodation and other flagged topics receive an explicit human-review message.

Other outcomes

A reply notice communicates other actionable results. Empty automatic checks stay quiet; this workflow has no resident-send path.

A positive opening still contains a grievance

Actual Telegram progression: Avery's check-in is sent, then the maintenance reply becomes an alert.



What entered

"We're happy here" is followed by a broken extractor fan and two prior reports.

What the automation did

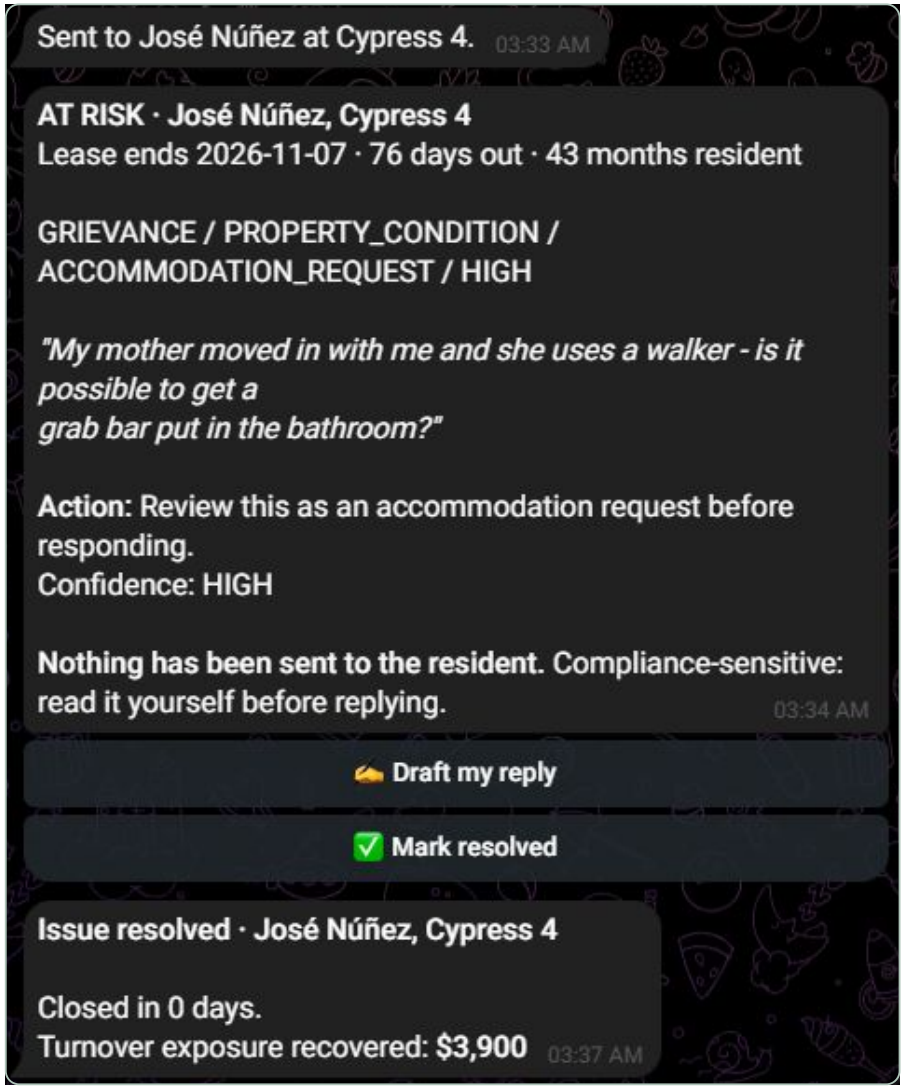
The recorded result is GRIEVANCE / MAINTENANCE / HIGH. It asks the manager to check work-order history and contact the resident.

What changed

Avery's Airtable status becomes AT_RISK. The manager can request a reply draft or record issue resolution.

An accommodation request goes to a person

Actual Telegram conversation / synthetic José Núñez at Cypress 4 / reference R-0545.



Sensitive input

The resident scenario asks about a bathroom grab bar for a family member using a walker.

Human review

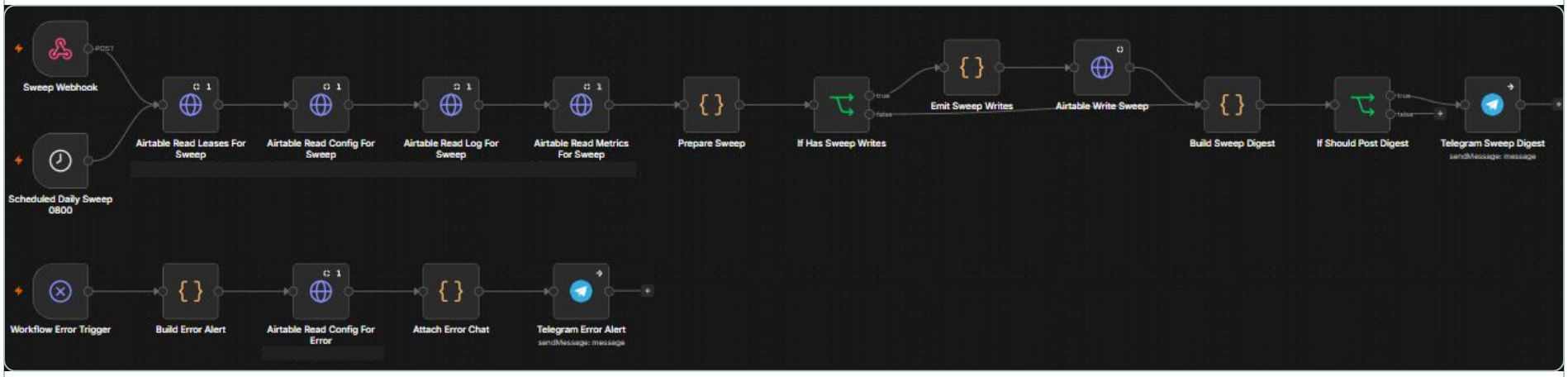
The alert includes ACCOMMODATION_REQUEST and states that nothing has been sent to the resident.

Manager-recorded closure

Mark resolved produces the closure card and an Airtable RESOLVED status. This demonstrates recorded issue closure, not a physical repair or a signed renewal.

Keep timing and metrics current

Actual n8n editor capture after Tidy Up. All 18 nodes are visible; enlarged branches follow.



FULL WORKFLOW / DARK THEME

Trigger and input

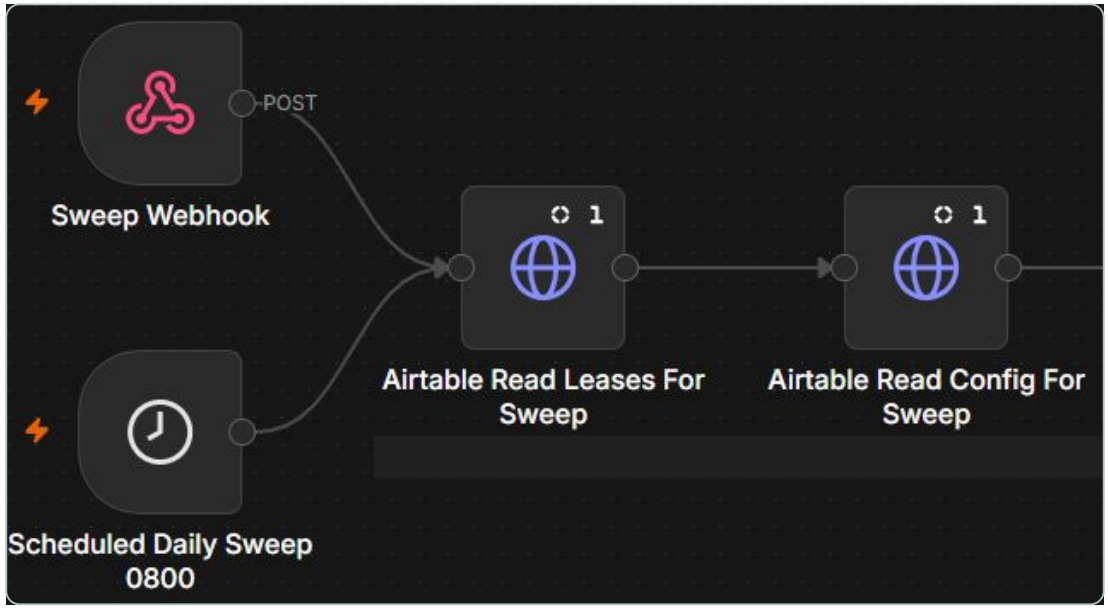
The daily 08:00 schedule, a sweep request or an internal refresh starts the main branch. It reads leases, configuration, outreach events and current metrics.

Key steps and output

Fixed rules recompute date cohorts and flag prolonged silence; the metric totals come from state and events. Airtable receives updated fields and the metrics snapshot. An optional Telegram digest reports results; a separate branch handles workflow errors.

Two triggers use the same sweep rules

Native-zoom captures of the triggers, data reads and preparation step.



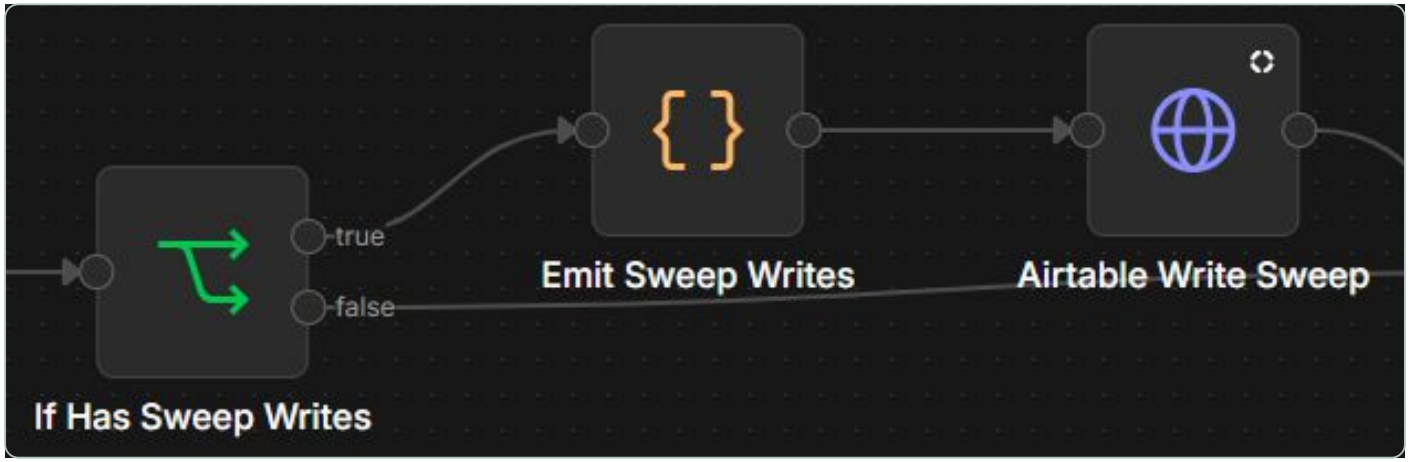
A schedule or sweep request joins the same Airtable read chain.



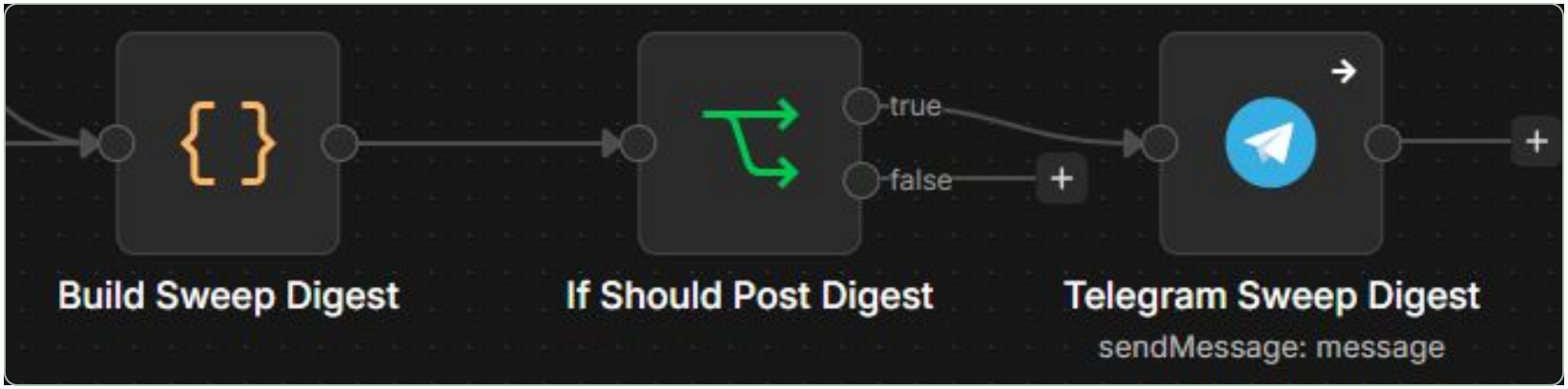
Prepare Sweep uses lease dates and logged activity to prepare the cohort, escalation and metric updates.

Refresh state, then summarize it

This workflow updates the manager's view without sending a resident message.



If Has Sweep Writes routes prepared changes through Airtable Write Sweep.



Build Sweep Digest creates the summary. If Should Post Digest keeps internal silent refreshes from adding routine Telegram messages.

Workflow failures have an alert path

The independent error branch is visible in the actual operations canvas.



Trigger

Workflow Error Trigger receives a failure from workflows configured to use the operations error handler.

Data movement

Build Error Alert prepares the failure context. Airtable supplies the manager-chat setting, and Attach Error Chat prepares the Telegram destination.

Output

Telegram Error Alert surfaces the failure for intervention. Resident sending is not part of the sweep or error workflow.

Recorded behavior

The project verified sweep recomputation, metric refresh and digest output. The error branch shown here documents the implemented routing; it is not a fabricated failure demonstration.

The scoreboard reflects the recorded journey

Actual populated Airtable Metrics table, preserved from the synthetic demo.

# residents_in_window	# check_ins_sent	# replies_received
14	2	2

14 residents in the outreach window, 2 approved check-ins sent and 2 replies received.

# issues_found	# issues_resolved
2	1

2 issues found and 1 manager-recorded resolution.

How the totals are built

Send and reply counts come from outreach events. Open and resolved issue fields determine the issue totals.

Meaning of the exposure figure

The demo uses a configured $3,900 per issue to show open and resolved turnover exposure. The resolution card is a tracking proxy; it does not demonstrate realized savings or a renewal outcome.

Poll replies through the same processing workflow

Actual full canvas after Tidy Up. All three nodes are shown at native zoom.



Trigger and input

Every Minute starts a scheduled check. Build Check Request prepares a silent request to WF8.

Key step and output

Run Reply Check calls the shared reply-processing workflow. New messages follow the same matching, routing, logging and manager-alert steps as a manual check.

Recorded outcome

The preserved executions that processed Avery's maintenance reply and José's accommodation request both entered WF8 with silent checking enabled. New-reply alerts still appeared in Telegram; empty polls do not post routine notices.

Input, human action and output stay linked

Actual Airtable event log. Private delivery identifiers are excluded from this capture.

event_type	channel	direction	lease_id	tenant_name
DRAFTED	SYSTEM	INTERNAL	cedar-row-9-2026-11-11	Avery Brooks
SENT	EMAIL	OUTBOUND	cedar-row-9-2026-11-11	Avery Brooks
REPLY_RECEIVED	EMAIL	INBOUND	cedar-row-9-2026-11-11	Avery Brooks
ASSIST_SENT	EMAIL	OUTBOUND	cedar-row-9-2026-11-11	Avery Brooks
SENT	EMAIL	OUTBOUND	cypress-4-2026-11-07	José Núñez
REPLY_RECEIVED	EMAIL	INBOUND	cypress-4-2026-11-07	José Núñez
RESOLVED	SYSTEM	INTERNAL	cypress-4-2026-11-07	José Núñez

Avery / Cedar Row 9 / R-33F8

Edited draft → SENT → REPLY_RECEIVED → ASSIST_SENT. The Telegram and Gmail evidence shows the corresponding actions and wording.

José / Cypress 4 / R-0545

SENT → REPLY_RECEIVED → RESOLVED. The sensitive review and manager-recorded closure appear in the conversation.

What this demonstrates

Genuine integrations, individual send approval and visible state changes. All resident identities and scenarios shown are synthetic.